Chapter 4

Tooth wear

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INTRODUCTION

Tooth wear is a common clinical finding and may cause diagnostic and management challenges. The early signs of tooth wear are almost universally present in adults but the more severe forms probably affect around 10% of the population. Essential to planning the long-term care of patients with tooth wear is identifying the cause, when to start prevention and when to intervene with operative management, which in most cases involves advanced restorative techniques.

DEFINITIONS

- *Erosion*. The loss of tooth structure caused by acids and not involving bacterial fermentation is called erosion. Acids demineralize the outer surface of enamel or dentine, increasing its porosity. Optimum concentrations of fluoride and calcium may allow some remineralization; however, if another exposure to acid or other forms of tooth wear attack the surface layer, then tooth tissue will be lost.
- *Abrasion*. Abrasion is the mechanical wear of teeth from forces other than teeth.
- *Attrition*. Attrition is the wear of tooth against tooth. It is uncommon to observe the singular effects of erosion, abrasion or attrition and more commonly it is a combination.
- *Abfraction*. Laboratory studies using finite element analysis have hypothesized that a condition called abfraction might occur. This is based on the fact that excessive lateral occlusal forces on teeth cause microscopic flexure, leading to cyclic stress concentration in the cervical area with resultant loss of tooth tissue (Figure 4.1). Although some researchers claim an association between bruxism and abfraction lesions, this theoretical phenomenon has yet to be proven clinically. The lesions that result cannot be differentiated from those that occur as a result of cervical erosion and abrasion. The multifactorial nature of tooth wear will also hinder proof of this hypothesis clinically.

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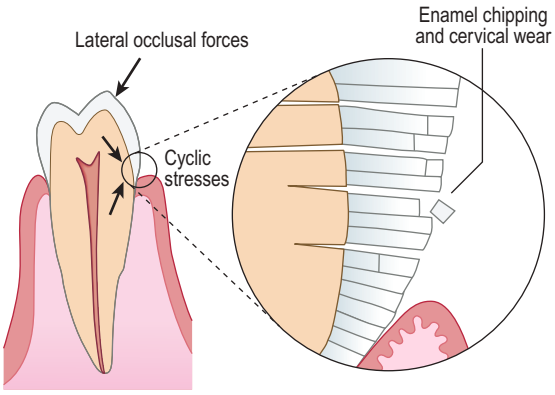


Figure 4.1 Diagrammatic representation of the stresses created at the cervical margin of a premolar tooth due to repeated lateral occlusal forces, leading to chipping and wear of the cervical area.



Figure 4.2 In the early stages of wear the anatomical features of the tooth become indistinct and features such as mamelons are worn away.



Figure 4.3 The appearance of erosive tooth wear on the palatal surfaces of the upper incisors. Note the exposed dentine centrally and the peripheral rim of enamel.

CLINICAL APPEARANCE

As tooth wear normally presents as a combination of erosion, abrasion and attrition, it is difficult to isolate the impact of the various components. However, there are clinical features which are specific to erosion, abrasion and attrition which can help with the diagnosis.

Anterior teeth

In the early stages of wear the anatomical features of the tooth become indistinct and lose features such as mamelons (Figure 4.2). Erosion results in a glassy appearance, particularly on the facial/buccal surfaces of the upper anterior teeth. As the condition deteriorates, the effects become more pronounced and widespread. Early stages involving enamel wear can be difficult to identify, particularly around the cervical margins of teeth. Localized and extensive lesions limited to enamel will be visible but generalized changes can sometimes be difficult to assess.

Once dentine is exposed, the difference in colour compared with enamel will make the diagnosis clearer. On the palatal surfaces of upper anterior teeth the complete loss of the enamel exposes the dentine, giving the characteristic appearance of perimolysis (Figure 4.3). This is a condition that can arise from chronic vomiting or regurgitation. It is also thought that the rough surface of the dorsum of the tongue may act as an acid reservoir and an abrasive, increasing the tooth wear and producing the characteristic highly 'polished' palatal surfaces with a rim of enamel (Figure 4.3). If the wear process proceeds, the gradual undermining and thinning of the incisal edges of the teeth causes fractures of the enamel and dentine and increases the translucency. The presence of incisal translucency is indicative of a late stage and signifies considerable tooth wear on the palatal surfaces of the upper anterior teeth (Figure 4.4).

Wear along the cervical surfaces of teeth is often mistakenly referred to as abrasion. However, the abrasive nature of modern toothpastes and toothbrushes makes abrasion unlikely. Mechanical wear will occur with overzealous brushing but the formation of V-shaped lesions along the cervical margin (Figure 4.5) is often related to a combination of abrasion and erosion. Historically, textbooks have shown images of rare causes of abrasion, such as that from pipe smoking or where seamstresses have used their teeth to cut thread. However, in today's society, there is a new threat that the dentist needs to be aware of: body piercings. Figure 4.6 shows the amount of tooth wear that has occurred due to the parafunctional chewing of a tongue piercing. Such piercings have also



Figure 4.4 A late stage effect of palatal tooth wear is incisal translucency seen from the labial surface. The loss of palatal enamel and dentine results in thinning of the tooth, increasing its translucency.



Figure 4.5 Cervical wear is normally found on the buccal surfaces of canines and premolars, producing these characteristic V-shaped lesions.

been the cause of other intraoral complications such as gingival recession.

Frequent acidic challenges may cause dentine sensitivity, particularly following cold stimuli. Dentine sensitivity results from fluid movement within the dentinal tubules which stimulates pulpal nerves – the so-called 'Brännström's hydrodynamic theory' of dentine sensitivity. Tubule fluid movement is caused by expansion and contraction of the fluid as a result of temperature changes or due to changes in osmotic pressure, typically with sweet foods. Normally the dentinal tubules are blocked by smear layer plugs which resist this fluid movement; however, frequent acid attacks can remove these plugs allowing fluid movement and dentine sensitivity.

The presence of staining on enamel or dentine can give an estimation of the progression of erosive tooth wear. A stain-free surface can suggest that any acid erosion is active whilst a stained surface suggests the process is inactive. The successful implementation of dietary controls will have an effect on the progression of erosive tooth wear, with the teeth becoming progressively more stained as the process becomes inactive (Figures 4.7 and 4.8).

Molar teeth

In the early stages discrete circular cavities may appear on the occlusal surfaces of molar and premolar teeth. These cavities, unlike caries, are hard and have a halo of whitened enamel surrounding the deeper yellow of dentine (Figure 4.9). The cause of these lesions is normally associated with erosion but some contribution from abrasion may also be possible. This is often referred to as 'cupping out' of the cusp tips, but may also be seen in relation to the incisal edges of the anterior teeth. If the process deteriorates, the discrete cavities join and the lesion increases in size. If the erosion is rapid and severe, the cusp tips or incisal edges of one or more opposing molar, premolar and/or anterior teeth may appear not to meet on closure into the intercuspal position (Figures 4.10 and 4.11) and, where present, gold, amalgam or composite restorations appear to be proud as the surrounding tooth structure wears at a greater rate than the restoration (Figure 4.12).

If the cause of the wear is predominantly attrition, the occlusal surfaces of either anterior or posterior teeth from



Figure 4.6 Parafunctional chewing of a tongue piercing (left) has led to wear of both upper and lower teeth (right).



Figure 4.7 The appearance of an unstained tooth surface is suggestive of an active erosive process occurring.



Figure 4.8 When erosion becomes inactive, the teeth become progressively more stained as dietary products accumulate on the tooth surface.



Figure 4.9 Small and discrete cavities formed on the occlusal surfaces of molar and premolar teeth are typically associated with erosion. There may also be an abrasive component, often referred to as 'cupping out' of the cusp tips.



Figure 4.10 A severe form of tooth wear with a predominantly erosive component resulting in the loss of occlusal contact with the opposing teeth. Note only the most posterior teeth are in contact.



Figure 4.11 When acid erosion is predominant the loss of occlusal or incisal tooth structure will prevent contact between the opposing teeth.



Figure 4.12 The gold restoration on the lower right central incisor tooth now stands proud following a greater differential wear of tooth tissue as a result of erosion and attrition.



Figure 4.13 Attrition is the wear caused by action of tooth against tooth, and the matched flattening of the incisal edges seen here is characteristic.

both dental arches demonstrate paired wear facets, but in more severe cases they may appear flattened, almost as if they have been filed (Figure 4.13) and the opposing cusps or incisal edges meet and closely interdigitate in the intercuspal position. Typically in severe cases fractured restorations are found following the excessive occlusal loading from the opposing teeth (Figure 4.14).

Sensitivity

Dentine sensitivity is a sign of acid erosion and normally occurs in patients with meticulous oral hygiene. The symptoms are a sharp, short duration pain or sensitivity when cold is applied to the teeth. It normally is most common along the cervical margins of teeth and frequently occurs in association with cervical wear. The aetiology is almost certainly a combination of erosion and abrasion but the role of acids is often ignored. Dentine sensitivity is relatively uncommon on surfaces other than the cervical areas. This is thought to be related to the formation of a smear layer sealing the dentinal tubules, whereas along the cervical margin the frequent removal of the smear layer by tooth brushing increases the risk of sensitivity developing. There are no signs or symptoms to indicate that tooth wear is active apart from the presence or absence of stains on the teeth. Since dentine sensitivity is thought to be an indicator of activity, if it is present alongside a clean, unstained tooth surface particular care should be taken when monitoring progression.



Figure 4.14 The effect of bruxism on the occlusal surfaces of premolar and molar teeth has caused a gradual flattening of the teeth. Often the high occlusal loading causes fracture of restorations which can be seen here on both direct and indirect restorations.

AETIOLOGY

Erosion

The most common cause of erosion is the frequent consumption of acids (extrinsic acids). The strength of the dietary acid is also important. Strong acids with low pH and high titratability will cause demineralization of the hydroxyapatite in the teeth (Table 4.1). The titratability is defined as the volume of alkali needed to neutralize an acid. Common examples of foods which contain strong dietary acids are citrus fruits. Whilst the intake of dietary acids is recognized as being potentially erosive, probably the most important component is the frequency of consumption. Some dietary habits such as sipping acidic drinks or snacking on fruits result in prolonged drops in oral pH which, when combined with other forms of wear such as attrition or abrasion, cause tooth wear. There is an increasing body of evidence to implicate dietary acids (particularly strong acids) in prevalence studies on tooth wear.

Table 4.1 Common dietary acids associated with acid erosion

EROSIVE POTENTIAL	DIETARY SOURCE
High erosive potential (fruit or fruit juices)	Citrus fruits (lemons, grapefruit, oranges, limes) Apples Cranberries Grapes
Medium erosive potential	Cola drinks Vinegar White and red wine
Low erosive potential	Beer Carbonated water



Figure 4.15 The appearance of palatal dental wear caused by gastric regurgitation is typically seen in patients with eating disorders and reflux.

The other major cause of erosion is the effect of gastric acids (intrinsic acids) associated with eating disorders or regurgitation. Anorexia and bulimia nervosa are relatively uncommon disorders predominantly affecting females between the ages of 13 and 20 years old. Sufferers use dietary restriction or vomiting to control their body image. Anorexia is more commonly associated with a low body weight and has other health-related problems such as amenorrhoea, low body mass and low metabolic rates to name but a few, whilst bulimics have a normal body weight but use binge eating and vomiting to control their weight. The distinction between the two conditions may merge, with some anorexics passing through bulimic phases and bulimics entering anorexic periods. The outcome on teeth is palatal dental erosion which is often severe (Figure 4.15).

Regurgitation of gastric juice resulting from an underlying gastro-oesophageal reflux condition is the other major cause of gastric acid passing into the mouth. In some situations the symptoms of reflux (heartburn, epigastric pain, regurgitation) are present but in around 25% there are no symptoms. The outcome on the teeth has an identical appearance to that of patients with an eating disorder. Irrespective of the cause of gastric juice entering the mouth, the outcome on the teeth is generally more severe than that found with dietary erosion. Fortunately, gastric causes of erosion are relatively uncommon but generally the severe cases result in the need for specialist prosthodontic care.

Abrasion

Although the cause of cervical tooth wear has been traditionally associated with overzealous tooth brushing (see Figure 4.5), most of the laboratory studies have shown that normal forces with tooth brushing, with or without toothpastes, is unlikely to cause significant wear on enamel or dentine. However, if excessive force is applied the mechanical action from the tooth brushing may result in wear and this normally appears on the cervical margins of premolar and canine teeth. Concurrent dentine sensitivity can indicate an erosive component to the wear and is successfully managed with dietary advice, together with non-traumatic tooth brushing instruction. The importance of abfraction along the cervical area, predisposing the surface to erosion or abrasion, is unknown.

Attrition

Wear from tooth to tooth contact is a common finding, with facets appearing on teeth used in guidance (e.g. canines and premolars). More severe forms of attrition occur with habits such as bruxism where continual and prolonged loading of teeth causes significant occlusal wear (see Figure 4.13). The reason for a bruxism habit is unknown but some component of stress causing repeated tooth clenching or grinding is acknowledged by most clinicians (see Chapter 6). Others suggest occlusal interferences may trigger the grinding or clenching habits, but since interferences are so common in asymptomatic subjects this association remains unproven.

MANAGEMENT OF TOOTH WEAR

Early detection and monitoring

There is no justification to believe that tooth wear inevitably leads to the total destruction of teeth. In some individuals the extent of the tooth wear may compromise the longevity of the tooth but in most the progression is slow and is part of the ageing process. This is an important factor as the severity of the tooth wear must always be balanced against the patient's age. For example, a degree of tooth wear seen in a 70-year-old patient may be regarded as normal 'wear and tear'; however, if the same amount of wear was seen in a 20-year-old patient, alarm bells might sound.

As with dental caries, the most important aspect to the management of tooth wear is early detection and identification of risk factors. Reducing the effect of these risk factors, which have been discussed previously in this chapter, can prevent further progression of the wear. For many patients whose tooth wear has been identified by their dentist and managed in this manner, monitoring is an effective and acceptable procedure even though there is no attempt to restore the shape and appearance of the teeth. Other patients may present with specific complaints or concerns in relation to the wear of their teeth and it is important to identify what these are and address them. The main issues that arise from tooth wear are:

- Poor appearance
- Poor function
- Sensitivity
- Concerns about continued wear.

One of the main challenges with restoring teeth with wear, which both dentist and patient have to be aware of, is the immediate and long-term cost of care and maintenance. With this in mind most patients presenting with concerns over continued tooth wear will be happy to accept preventive advice and monitoring; the reassurance given that the condition is not deteriorating with time is all that they require. Indeed, many of those concerned about appearance and function will also reflect and accept this advice as the restoration of worn teeth is normally expensive and can require specialist levels of care, mainly because it rarely affects single teeth and more commonly involves sextants, quadrants or even entire arches.

Sensitivity can usually be treated with minimally invasive techniques and changes in dietary habits. Products and techniques used for the management of dentine sensitivity aim to either block the dentinal tubules and so prevent fluid movement or interfere with neural transmission within the pulp. Dentinal tubules can be blocked with agents in toothpaste such as strontium chloride, calcium phosphate or oxalates, or professionally applied dentine bonding agents or adhesive restorations. Topical fluoride application may also promote remineralization within the dentinal tubules. Neural transmission may be affected by potassium salts; potassium is thought to diffuse up the dentinal tubules, altering the membrane potential of pulpal nerves, so decreasing their excitability and neural transmission.



Figure 4.16 Early enamel wear may be predictive of future progression.



Figure 4.17 Exposure of dentine does not necessarily mean that the wear will continue at the same rate throughout life.

Preventive management

There is some evidence to suggest that if preventive advice is given the rate of tooth wear will reduce and further wear becomes consistent with the ageing process. The role of prevention is often overlooked in favour of operative intervention, but monitoring and having a long-term view on the prognosis of worn teeth is a valuable management technique. Whether the presence of early enamel wear is an indicator of later more severe wear is not known. Prevention of wear, however, particularly that associated with acids, should be considered when the effects of wear are first recognized (Figures 4.16 and 4.17).

The most important aspect to prevention is monitoring the rate of the tooth wear. This is traditionally accomplished by examining sequential study casts or comparing one set, with the date marked, to the patient's teeth. Often it is easier to compare like with like; if sequential study casts are used, an accurate and consistent impression technique is required to record the fine detail of wear (see Chapter 13). As damage to study casts is not uncommon, especially if kept for prolonged periods, it is worthwhile requesting that your technician cast these up in more durable die stone. In a busy clinic storage of models can be problematic and patients often move location. For these reasons it is sensible to give the dated models to the patient to bring with them at subsequent review appointments, taking care to document this in the patient's notes. Some clinicians use matrices to assess how much the wear has progressed but perhaps the simplest method is to use a periodontal probe and measure the height or width of the buccal, incisal/occlusal and palatal tooth surfaces (Figures 4.18 and 4.19).

Dietary advice

If extrinsic acids are thought to be the primary cause of the wear, a diet diary should be carried out like that for dental caries management. Acid content and frequency of intake should be identified and realistic alternatives suggested; for example, replacing juice drinks with water. Eating and drinking habits should also be discussed; for



Figure 4.18 Monitoring of tooth wear can be effective. If the rate of tooth wear diminishes and there are undetectable changes, then monitoring can continue.



Figure 4.19 The original study cast taken for the patient in Figure 4.18. Comparison to the teeth shows no change over 15 years.

example, holding or swilling drinks in the palatal vault prior to swallowing can exacerbate erosion. Advice on the use of a straw placed toward the back of the mouth and avoidance of bathing the teeth in the beverage should be given if the patient cannot completely cease consumption. Acid drinks should also be avoided last thing at night and patients should be advised not to brush their teeth immediately following an acid drink, but rather rinse with water to clear and dilute the acid.

Fluoride

The role of topical fluoride in the preventive management of dental caries is unequivocally recognized as an effective and efficient modality. The strength of the evidence to support its use in tooth wear is not as well established but the evidence is mounting. There is a good level and varied supply of laboratory evidence to show that fluoride will help to prevent erosion and abrasion. Further evidence, mainly from in situ studies, also supports the preventive action of fluoride; however, there is virtually no clinical patient-based evidence. Nevertheless, based on the current evidence, fluoride is protective for erosive tooth wear.

Fluoride is thought to harden the enamel surface, increasing its resistance to acid dissolution. As acids erode the outer surface of enamel, some penetration of the surface layer is inevitable. This seepage through the enamel surface partly demineralizes and increases the porosity of the tissue, possibly reducing its resistance to further wear. However, if fluoride is freely available it can harden the outer layer and assist in the remineralization of the deeper ones. The effectiveness of fluoride on dentine is understood less but there may be a similar mechanism.

Saliva

The role of saliva in erosion is unclear. Theoretically, an absence should lead to an increase in the severity of erosion but this is not always seen clinically. The reason for this dilemma is not completely understood. It might be that xerostomia results in the severe hypersensitivity to acidic drinks and therefore dietary avoidance. The data from studies comparing saliva to prevalence of tooth wear or erosion are also equivocal, with some studies showing a positive correlation and others not. In principle, the role of saliva should be important in the development of erosive tooth wear.

Dentine bonding agents

Sealant restorations in caries management are well established but in erosive tooth wear there is limited evidence. Using dentine bonding agents to prevent dentine sensitivity is also well established and part of routine care. It is therefore likely that using dentine bonding agents to seal the early erosive lesions would also be effective in prevention of wear. There is some clinical evidence to support the role of dentine bonding agents and their use should be considered in patients where sensitivity is a problem or when ineffective control of the risk factors occurs.

Clinical and laboratory studies have shown that dentine bonding agents can reduce the amount of wear by half over a period of 6 months. However, the bonding agents are thought to be gradually worn away by mechanical action and repeated applications at regular review, particularly on susceptible tooth surfaces, is a preventive and minimally invasive technique.

Splint therapy

The use of full coverage splints will be effective for those patients with mainly a bruxist habit. Their use in patients where erosion is a major contributory factor is probably

ineffective. The splints used to prevent bruxism can be made in either a vacuum-formed polyvinyl material or in heat-cured acrylic. Although neither has been shown to be more effective, the durability and longevity of the latter will be better. The use of splints with mandibular dysfunction is a well-recognized treatment but is often forgotten with bruxism and attrition (see Chapter 6) in the haste to provide more interventive restorative treatment.

The use of splints, particularly for those patients with bruxism, can be very effective as crowns and restorations often break under the extreme occlusal loading caused by the habit. A full coverage hard acrylic splint is relatively straightforward to make and simple to fit. Upper and lower impressions should be taken in a silicone material unless an alginate can be poured within 30 minutes. The silicone impressions should be sent to the laboratory with an interocclusal record and a request to make a full coverage upper hard acrylic splint with canine rises in lateral excursions. This is commonly referred to as a 'Michigan splint' (Figure 4.20). On receiving the splint from the laboratory it should be adjusted to ensure that there are even occlusal contacts in the intercuspal position and there is bilateral canine guidance. Providing the laboratory with a facebow recording for mounting of the models on a semi-adjustable articulator will reduce the amount of occlusal adjustment and hence clinical time needed to fit the splint.



Figure 4.20 A Michigan full coverage maxillary splint used to protect teeth from the effects of bruxism. Normally worn at night, the splint distributes the loading of the clenching or grinding action and protects the teeth and restorations from cracking and fracturing.

When to restore?

The judgement on whether or not to restore worn teeth is in part related to the expectations and wishes of the patient but also to some extent on whether or not the teeth might survive for a lifetime. The main indications which eventually demand restorative treatment of worn teeth are appearance and intractable sensitivity. Another often quoted indication is whether or not the rate of tooth wear will compromise the longevity of the teeth. This judgement is by its very nature an estimate and relies on clinical experience. However, so little is known about the rate of tooth wear and the impact of prevention that the real outcome can never be known. There is a suspicion that tooth wear varies in activity with periods of progression and quiescence. One important assessment in the management of worn teeth, particularly those severely affected by wear, is the ability of the patient to afford the cost of care and maintenance. Since severe wear normally affects more than one tooth, the cost to restore them is high. Whether or not the patient wishes to improve their appearance, if they cannot afford complex care and aftercare, then monitoring with intensive preventive management may be an effective outcome.

SUMMARY

Tooth wear is a multifactorial, complex process that can lead to dramatic defects and loss of tooth structure. Early detection, preventive advice (and treatment) and monitoring the outcome of this form of management is of paramount importance. The need for advanced operative techniques may therefore be avoided. In those patients that present with a late stage of wear or where preventive management fails and there is evidence of continued wear from dated study casts, operative treatment may be indicated. It is important that prior to taking this decision the patient is aware of the complexity and long-term implications of such management and that this is clearly documented in the patient's notes. The operative management of patients with tooth wear will be discussed in Chapter 16.

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